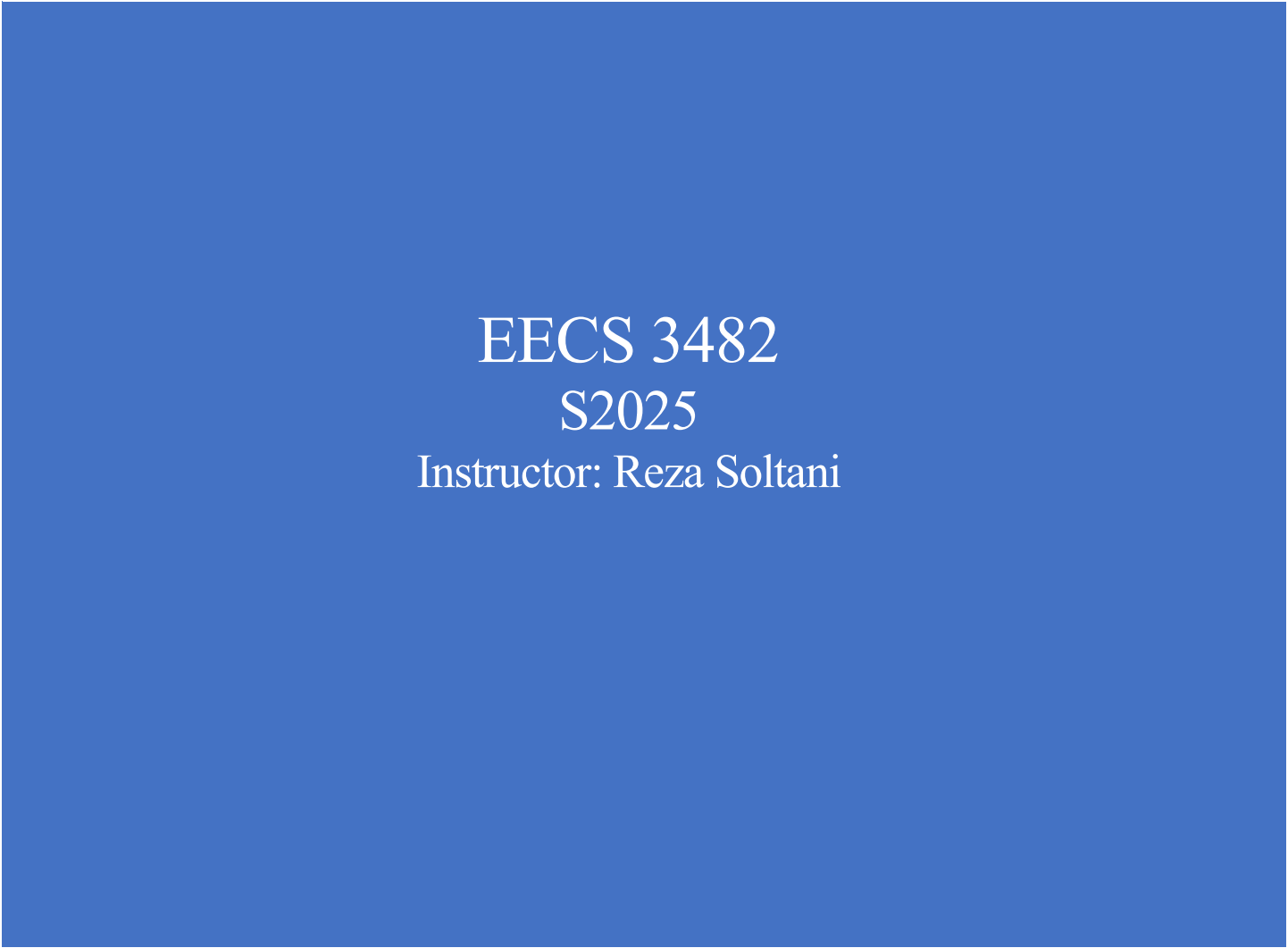




Lab 2 – Linux Operations and Social Engineering



EECS 3482
S2025
Instructor: Reza Soltani

The purpose of this lab is to become familiar with basics of Linux operations and to perform Social Engineering attacks.

General Information and Policies

- Solutions must to be submitted through the eClass system.
- Reports should include screen snapshots of all of the steps as well as copies of data or script segments where applicable.

Objectives

- Sample and test social engineering approaches.
- Review operating systems commands useful for credentials retrievals

Deliverables

Lab Report, including screen snapshots, scripts, codes, and output of Part-1 to Part-2

Lab Format

For every step of your labs, including this lab and all future labs in the course, you must provide a clear and **legible screenshot** whenever you use the command prompt.

Before taking the screenshot, and either before or after completing the step, you **must** enter the following command so that your **username**, **full name**, and **timestamp** are visible in the screenshot:

```
echo " Step completed by $(whoami), name: <first> <last>, on $(date)"
```

Note, replace *<first>* and *<last>* with your own first name and last name before running the command.

For example, a student with the name *John Smith* and username *JSUser1* should run the following command:

```
echo "Step completed by $(whoami), name: john smith, on $(date)"
```

to get the following result:

```
Step completed by JSUser1, name: John Smith, on Sat 17 May 2025 12:48:05 EDT
```

Note, there are different kinds of shell interface within Unix/Linux systems. Depending on the environment you are in (such as the *eeecsvm*'s kali environment which uses **ksh** instead of **bash** shell), you may have to enter the following modified command:

```
echo " Step completed by `whoami`, name: <first> <last>, on `date`"
```

or alternatively, enter **bash** in the command prompt to switch to *bash* shell before entering the original *echo* command.

Also note: If you are working on a Windows machine, you must use an equivalent command such as:

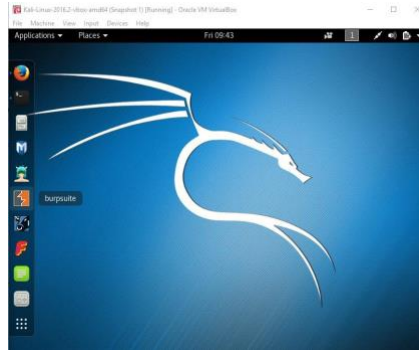
```
echo Step completed by %USERNAME%, name: <first> <last>, on %DATE% %TIME%
```

If a lab step does not include the above command or a clear screenshot, marks will be deducted.

Part 1: Wargames - AntiBandit

- Access the resource: <http://overthewire.org/wargames/bandit/> and play the **Bandit** game.
- Explore and complete the game stages from **Level 0** to **Level 13**. Document your every step of every level by describing the input and output.
- Include screenshots for your steps and make sure to include your name and date in them.
- Note, you do not need to do Level 13 -> Level 14.

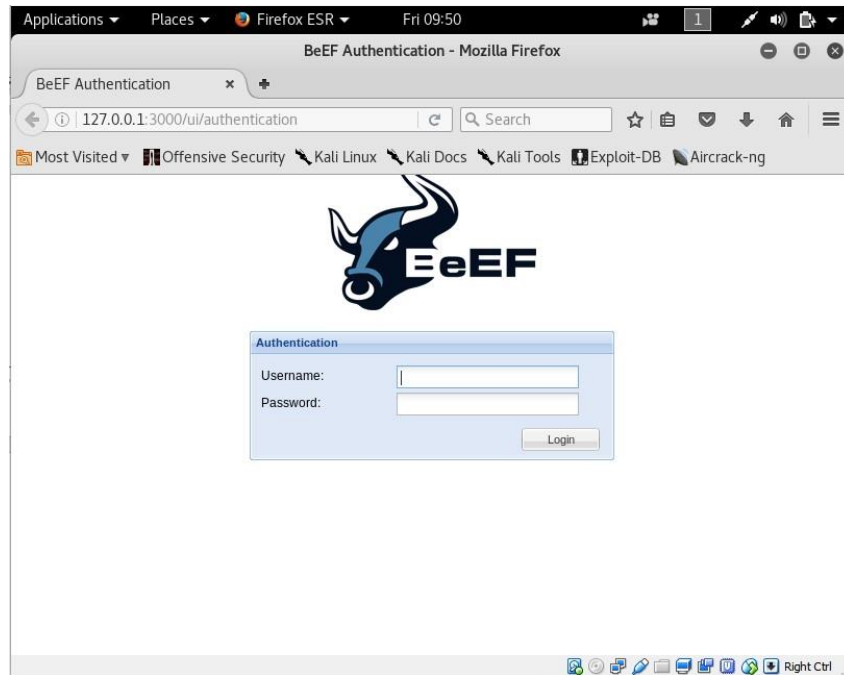
Part 2 - Kali BeEF XSS Framework



BeEF, short for Browser Exploitation Framework, is a penetration testing tool designed to target web browsers. It works by using social engineering tactics to "hook" a user's browser, allowing the attacker to remotely interact with it through BeEF's control panel. This framework highlights how even a fully updated and patched operating system can still be compromised through the browser. BeEF is commonly used in security assessments to demonstrate the risks of client-side attacks and to show how easily users can be exploited through deceptive tactics like fake login pages or malicious pop-ups.

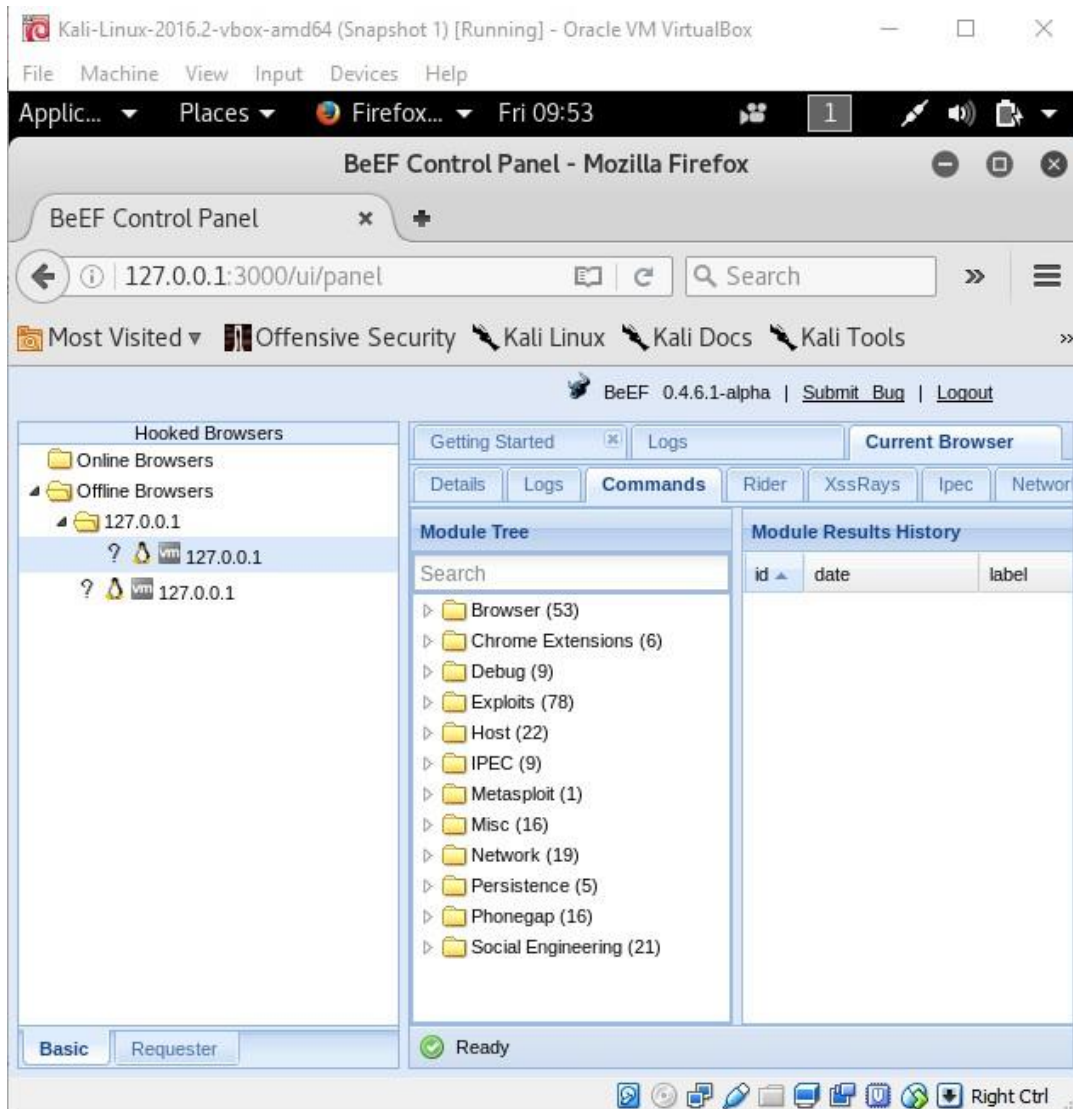
Follow the steps below:

1. From Kali, run the [BeEF XSS Framework](#)
2. If BeEF is not installed or need to be updated, follow the steps under the Notes section.
3. Login to the framework using the user "beef" and password "beef", or the username and password you used during installation.



4. You should hook to a browser by clicking on "... the advanced version [here](#).", a new page should be browser tab should be created that has the URL:
http://127.0.0.1:3000/demos/butcher/index.html
- Document the page using a screen snapshot (The Butcher page).

5. Click on the <http://127.0.0.1:3000/ui/panel>, which is BeEF control panel tab (the URL is: <http://127.0.0.1:3000/ui/panel>)
6. Double click on the hooked browser 127.0.0.1 to access the framework menu



5.1 Click on Commands

- 5.1.1 Select the Social Engineering folder
- 5.1.2 Select the Clippy vulnerability
- 5.1.3 Click ➔ Execute
- 5.1.4 Open the Butcher page and document the output using a screen snapshot.

5.2 Click Commands ➔ Select the Social Engineering folder

- 5.2.1 select “Fake LastPass” (Requires Chrome) or Fake Notification Bar (Firefox)
- 5.2.2 Click ➔ Execute
- 5.2.3 Open the Butcher page and document the output using a screen snapshot.

5.3 Click Commands ➔

- 5.3.1 Find WordPress Login or WordPress Add User, or any Wordpress related command
- 5.3.2 Click ➔ Execute
- 5.3.3 Open the Butcher page and document the output using a screen snapshot.

5.4 Click Commands ➔ Google Phishing

- 5.4.1 Click ➔ Execute
- 5.4.2 Open the Butcher page (URL: <http://127.0.0.1:3000/demos/butcher/index.html>) and document the output using a screen snapshot.

5.5 Click Commands ➔ Pretty Theft

- 5.5.1 Click ➔ Execute
- 5.5.2 Open the Butcher page (URL: <http://127.0.0.1:3000/demos/butcher/index.html>) and document the output using a screen snapshot.

5.6 Find and execute the ‘Browser Fingerprint’ command. Then, similar to the previous tasks, produce a screenshot from the Butcher page.

Note, if any of the Commands, is not available for your version of BeAF, then you can try a different command. Make sure to document the name of the command and its result.

What to Submit?

- Make sure you document all of your steps for part 1 and 2, and Include everything in a single PDF, with proper numbering and sections. Make sure your screenshots include your name and date.
- Submit your PDF before the deadline through eClass!

Notes about BeEF

In case you need to update/install BeEF on your machine:

Update your Linux apt:

- `sudo apt upgrade -y`
- `sudo apt dist-upgrade -y`

If the above fails, try:

- `sudo apt-get update`

Install BeEF:

`sudo apt install beef-xss`

Note: the password for *sudo Kali* is 'Kali' which is the same password you use to login to Kali.

If running into any issues with managing the password for BEEF, check:

<https://github.com/beefproject/beef/issues/1763#issuecomment-562286867>

You can learn more about BeAF here: <https://beefproject.com/>